

Sunny

Adaptive Real-Time Voice Companion



A voice agent that responds to what you say - and how you naturally take turns.

AssemblyAI Voice Agent Hackathon

One voice flow. Distinct responsibilities.

AssemblyAI perceives speech; Sunny handles the turn and playback.

Physical mic -> AssemblyAI SpeechStarted / Turn -> Sunny -> speaker

New human speech during playback triggers Sunny playback stop.

The accepted new turn then produces a new Sunny response.

Human E2E and true barge-in: verified existing physical-device sessions.

The public browser demo is a visualization, not an online backend.

What changes across observations

Existing offline RhythmAdapter source; synthetic replay, not a human benchmark.

record_user_pause -> rolling mean -> bounded next_pause_ms

Synthetic pause input: 900 ms 900 ms

Next-response wait policy: 300 -> 360 -> 414 ms

Original simulation consumes this value before its next response.

Not wired into the human-tested live harness; no persistence claim.

Public source excerpt + source hashes + executable tests: docs/learning-proof.md

Real microphone test - sanitized proof

Human E2E: 20260928_184302 | Human barge-in: 20260928_184855

0.000000 s	Playback start
10.345000 s	AssemblyAI SpeechStarted
10.345000 s	PlaybackStop trigger
12.351398 s	TurnFinal / new turn
14.744950 s	New response playback

Relative event times; identical timestamps do not prove zero acoustic latency.

No human audio, transcript or captured waveform is public. Not literal tester footage.

A narrow, verifiable adaptation claim

Numerical pause state is public; identity, personality and emotion inference are out of scope.

VERIFIED: state updates in the existing offline pause-policy prototype.

VERIFIED: separate human microphone and barge-in sessions.

NOT VERIFIED: live automatic learning or cross-session preferences.

OMITTED: JEF feedback-loop claim; existing Jev is a rule-based router.

OMITTED: relationship-aware preference distillation (future work).

For Judges - verify Sunny in 60 seconds

Reproduce the numerical proof; inspect the human event evidence.

1. README: responsibility boundary and direct evidence links.
2. docs/learning-proof.md: unchanged source methods and integration limits.
3. docs/human-validation.md: recorded human event ordering.
4. Run: `python -m unittest discover -s tests -v`
5. Final film 01:05-01:20: sanitized real-microphone event proof.

github.com/rickytzai/sunny-mi087